IS 3001

Sampling Techniques

Group Project – Group 14

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INTRODUCTION

The US Health Insurance Dataset contains information about individuals' health and demographic characteristics along with their medical insurance charges. This dataset contains 1338 rows of insured data, where the Insurance charges are given against the following attributes of the insured: Age, Sex, BMI, Number of Children, Smoker and Region.

This study is based on data from the following variables.

- **Age -** The Age variable represents the age of the insured person in years and is measured as a continuous numerical variable.
- **Sex -** The Sex variable indicates the gender of the insured person and is categorized as either male or female.
- **BMI -** The BMI (Body Mass Index) variable is a continuous numerical measure calculated as weight in kilograms divided by the square of height in meters (kg/m^2). It serves as an indicator of body fat.
- **Children -** The Children variable represents the number of children or dependents covered under the insurance plan. It is a numerical variable.
- **Smoker -** This is a categorical variable. The Smoker variable identifies whether the insured person is a smoker, with possible responses being yes or no.
- **Region -** The Region variable specifies the residential area of the insured person within the United States, categorized into northeast, northwest, southeast, and southwest. It is a categorical variable.
- **Charges -** This is a numerical variable. The Charges variable represents the individual medical insurance cost billed by the insurance company in U.S. dollars.

OBJECTIVES

- To select samples using simple random, stratified, and two-stage cluster sampling methods.
- To estimate the mean, total, proportion and standard errors for selected variables under each sampling design.
- To present graphical analyses for the selected variables.
- To identify the most suitable sampling method for this dataset.

METHODOLOGY

According to the selected dataset, there are 1,338 individuals in the population. We applied different sampling designs to identify the most appropriate method for this study. The sampling designs used were Simple Random Sampling, Stratified Sampling, and Two-Stage Cluster Sampling.

For better accuracy and comparison, two samples were obtained under each design. First, we calculated the required sample size for each sampling design. Then, we selected the most suitable variables for stratification and clustering. After selecting the stratified and cluster variables, sampling weights were calculated for each design to ensure that every sampled observation correctly represented its proportion in the total population. Then, we identified the most appropriate variables and estimated the mean, proportion, and their standard errors under each sampling design.

Subsequently, we calculated the actual population values from the dataset and compared them with the estimates obtained from each sampling design. Based on these comparisons, we determined the most suitable sampling design for this study. Finally, a brief graphical analysis was carried out for the selected variables to visualize the differences and validate the findings. All analyses and calculations were performed using the R Software Package.

SAMPLING DESIGNS

i. Simple random Sampling

- A method of selecting a subset from a population where every individual has an equal probability of inclusion. In our sampling design, we applied simple random sampling without replacement to ensure unbiased representation**.**

ii. Stratified Sampling

- In this sampling method, the entire population is divided into non-overlapping homogeneous subgroups called strata. Each sampling unit belongs to one and only one stratum, and a simple random sample (SRS) is then selected from each stratum to ensure proper representation of the whole population.

iii. Two-Stage Cluster Sampling

- A sampling design in which the population is first divided into clusters, and a random sample of these clusters is selected in the first stage. In the second stage, a simple random sample of elements is drawn from each of the selected clusters. This design is often used when a complete list of all population elements is difficult to obtain, but a list of clusters is readily available.

- These are some key points about two-stage cluster sampling.

  i. The population is divided into **clusters**, which are natural groups of elements
  ii. **Stage 1:** Randomly select a number of clusters from the population.
  iii. **Stage 2:** From each chosen cluster, take a **simple random sample** (SRS) of individual elements.
  iv. Useful when it is **costly or impractical** to list every element of the population.
  v. Provides **greater efficiency** in data collection, especially for large or geographically spread populations.
  vi. Can lead to **higher sampling error** compared to simple random sampling if clusters are not homogeneous.

SELECTING VARIABLES FOR STRATIFIED SAMPLING AND CLUSTER SAMPLING

❖ Selecting Variables For Stratified Sampling

The variable "smoker" was chosen as the stratification variable because smoking is a key lifestyle factor that has a strong impact on health status and medical expenses. Individuals who smoke generally have higher health risks and therefore tend to incur higher insurance charges compared to non-smokers. By stratifying the population based on smoking status, the study ensures that both smokers and non-smokers are properly represented in the sample.

❖ Selecting Variables For Cluster Sampling

The variable "region" was chosen as the cluster variable because it naturally divides the population into distinct geographical groups — northeast, northwest, southeast, and southwest. Each region can be considered a separate cluster since individuals living in the same area are likely to share similar living conditions, lifestyle patterns, and healthcare access. It allows comparisons between regions to identify possible regional differences in health insurance charges and related variables.

SAMPLE SIZE CALCULATION

1. Simple Random Sampling

   In Simple Random Sampling, every individual in the population has an equal chance of being selected.
   The sample size (n) can be calculated using the formula:

   $$\mathbf{n = no/(1+ no/N)} \quad \text{where } no = (Z_{\alpha/2} * S/e)^2$$

   we got 594 out of 1338 for our simple random sample size.

2. Stratified sampling

   In Stratified Sampling, the population is divided into strata based on a variable such as smoker. Then a sample is taken from each stratum. The Neyman allocation method is used to determine how many samples should be selected from each stratum to minimize the variance of the estimate.

   Substituting V for S^2 in the sample size (without fpc) formula of SRS, we can calculate the sample size for stratified sampling.

   $$no = (Z^2_{\alpha/2} * V) / e^2 \quad \text{where } V = \sum^{H}_{h=1}(n/n_h)(N_h/N)^2 S_h^2$$

   we got 594 as sample size for stratified sampling.

3. Two-stage cluster sampling

   In Cluster Sampling, the population is divided into groups using region variable. A few clusters are randomly selected and then a subset of individuals are sampled.

   We got 600 as sample size for cluster sampling.

SAMPLING WEIGHT

Sampling weights are used to ensure that each selected observation properly represents its portion of the total population.

$$\text{Sampling weight } (w_i) = 1/ P(\text{unit I will be included in a sample})$$

1. Simple Random Sampling

In Simple Random Sampling, every individual in the population has an equal chance of being selected.

$$w_i = N/n$$

2. Stratified Sampling

In Stratified Sampling, the population is divided into strata and samples are taken separately from each stratum.

Sampling weight for unit j of stratum h is,

$$w_{hj} = N_h/n_h$$

3.Two-Stage Cluster Sampling

In Cluster Sampling, the population is divided into clusters. Then clusters are selected and individuals are sampled within those clusters.

Sampling weight for ssu j of psu i is,

$$w_{ij} = NM_i / nm_i$$

POPULATION PARAMETERS

- The population's mean, total and proportion values were collected in order to contrast them to the estimated values for each design.
- Subsequently, the estimates obtained from each sampling design were compared with the actual population values to determine the most suitable design for the given dataset.

```
# Function to calculate true census mean and total for quantitative variables
for (var in quantitative_vars) {
  mean_val <- mean(census_data[[var]], na.rm = TRUE)
  total_val <- sum(census_data[[var]], na.rm = TRUE)
  pop[[var]] <- list(Mean = mean_val, Total = total_val)
}
```

```
# Calculate proportion of the first level
  proportion_val <- sum(census_data[[var]] == first_level, na.rm = TRUE)
  total_val <- sum(census_data[[var]] == first_level, na.rm = TRUE)
```

PARAMETER ESTIMATION

- The 'survey' package was utilized to compute the required estimates. Each sampling design was defined using the 'svydesign' function, and the variables were evaluated using “svymean” and “svytotal” functions.

simple random sampling

```
srs_design <- svydesign(
  ids = ~1,                # No clusters/PSUs
  fpc = rep(N, nrow(srs_sample)),     # Finite Population Correction (fpc = N, N, ...)
  data = srs_sample
)
```

stratified sampling

```
  strat_design <- svydesign(
    ids = ~1,
    strata = ~smoker,                                  # Stratification variable
    fpc = ~Nh,                        # The total population size N in each stratum
    data = strat_sample
  )
```

cluster sampling

```
cluster_design <- svydesign(
  ids = ~region,
  fpc = ~Ni,
  data = cluster_sample_data
 )
```

Mean

```
# Estimate Mean and SE
mean_est <- svymean(formula_mean, survey_design, na.rm = TRUE)
```

Total

```
# Estimate Total and SE (scaled to population size)
total_est <- svytotal(formula_mean, survey_design, na.rm = TRUE)
```

Proportion

```
# Estimate Proportion (Mean of dummy) and SE
prop_est <- svymean(formula_prop, survey_design, na.rm = TRUE)

# Estimate Total and SE
total_est <- svytotal(formula_prop, survey_design, na.rm = TRUE)
```

RATIO ESTIMATIONS

1. **Ratio estimation in SRS design**

$$\hat{B} = \frac{t_y}{t_x} \qquad \hat{t}_{yr} = \hat{B} * t_x$$

```
# Ratio Estimator Function
ratio_estimator <- function(sample_data, y_var = "charges", x_var = "bmi", pop_mean_x) {
  sample_data[[x_var]] <- as.numeric(sample_data[[x_var]])
  sample_data[[y_var]] <- as.numeric(sample_data[[y_var]])
 xbar <- mean(sample_data[[x_var]])
  ybar <- mean(sample_data[[y_var]])
  R_hat <- ybar / xbar
  Mean_Y_hat <- R_hat * pop_mean_x
  return(list(Ratio = R_hat, Est_Mean_Charges = Mean_Y_hat))
}
#PART 1 — SIMPLE RANDOM SAMPLING (SRS)
run_srs_ratio <- function(seed, label) {
  set.seed(seed)
  sample_srs <- census_data %>% sample_n(200)

 res <- ratio_estimator(sample_srs, "charges", "bmi", pop_mean_bmi)

  return(data.frame(
    Sample = label,
    Ratio_Estimate = round(res$Ratio, 4),
    Estimated_Mean_Charges = round(res$Est_Mean_Charges, 2)
  ))
```

```
}


srs1 <- run_srs_ratio(101, "SRS Sample 1")

srs2 <- run_srs_ratio(202, "SRS Sample 2")
```

2. **Ratio estimation in Stratified random sampling design.**

   - In stratified sampling, there are two methods for ratio estimation.

     i. Separate ratio estimator

     ii. combined ratio estimator

   - The separate ratio estimator performs efficiently when the ratio $(\hat{t}y/\hat{t}x)$ varies considerably across strata. In this dataset, the ratio does not show significant variation among strata, making the separate estimator less efficient. Moreover, some strata have relatively small sample sizes, which increases variability. Hence, the combined ratio estimator provides a more stable and less biased estimate. Therefore, the combined ratio estimator was chosen for the Stratified Sampling design.

```
# PART 2 — STRATIFIED SAMPLING (by smoker-10% from each stratum)
run_strat_ratio <- function(seed, label) {
  set.seed(seed)
  sample_strat <- census_data %>%
    group_by(smoker) %>%
    sample_frac(0.10) %>%
    ungroup()

  res <- ratio_estimator(sample_strat, "charges", "bmi", pop_mean_bmi)

  return(data.frame(
    Sample = label,
    Ratio_Estimate = round(res$Ratio, 4),
    Estimated_Mean_Charges = round(res$Est_Mean_Charges, 2)
  ))
}

strat1 <- run_strat_ratio(303, "Stratified Sample 1")
strat2 <- run_strat_ratio(404, "Stratified Sample 2")
```

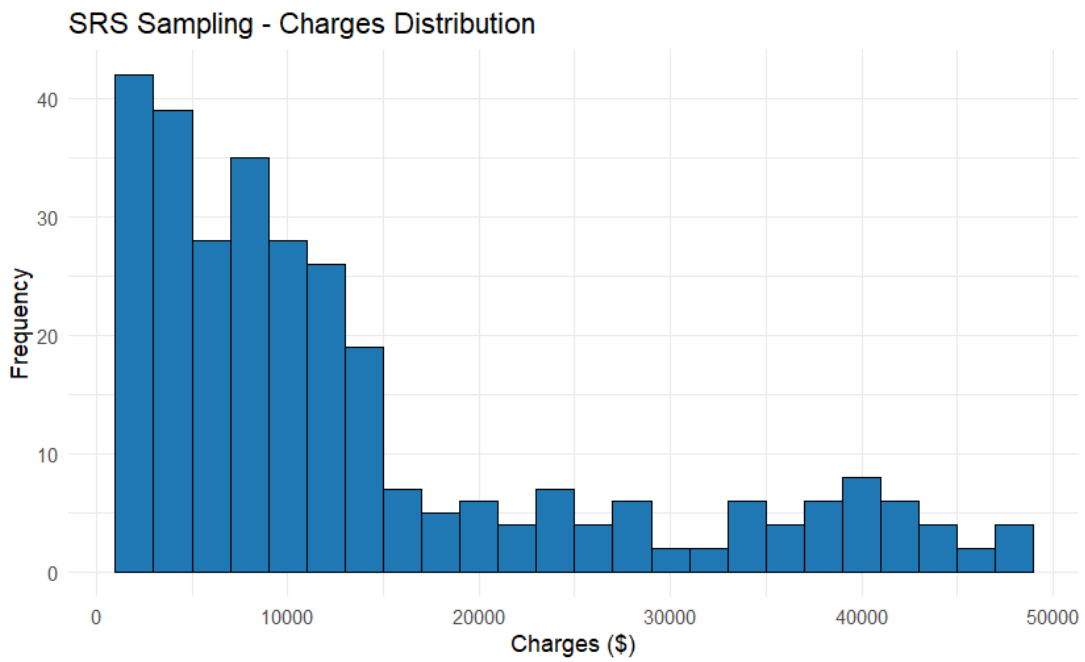
3. **Ratio estimation in Cluster sampling design.**

$$\hat{y}_r = \sum_{i \in s} \hat{t}_i \,/\, M_i \qquad \hat{y}_r = \sum_{i \in s} \hat{t}_i \,/\, M_i \qquad M_0 = \sum_{i \in s} M_i$$

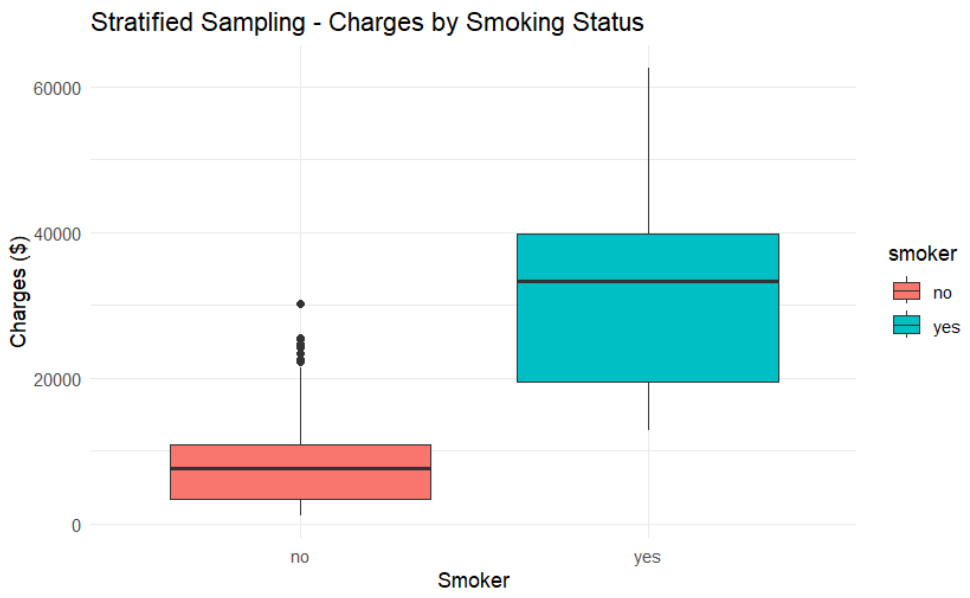
```
#PART 3 — TWO-STAGE CLUSTER SAMPLING (by region)
run_cluster_ratio <- function(seed, label) {
  set.seed(seed)
  # Stage 1: select 2 clusters
  selected_clusters <- sample(unique(census_data$region), 2)
   # Stage 2: sample equal number per cluster
  sample_cluster <- census_data %>%
    filter(region %in% selected_clusters) %>%
    group_by(region) %>%
    sample_n(50) %>%  # 50 from each selected region
    ungroup()
   res <- ratio_estimator(sample_cluster, "charges", "bmi", pop_mean_bmi)
  return(data.frame(
    Sample = label,
    Ratio_Estimate = round(res$Ratio, 4),
    Estimated_Mean_Charges = round(res$Est_Mean_Charges, 2)
  ))
}

cluster1 <- run_cluster_ratio(505, "Cluster Sample 1")
cluster2 <- run_cluster_ratio(606, "Cluster Sample 2")
```

GRAPHICAL ANALYSIS

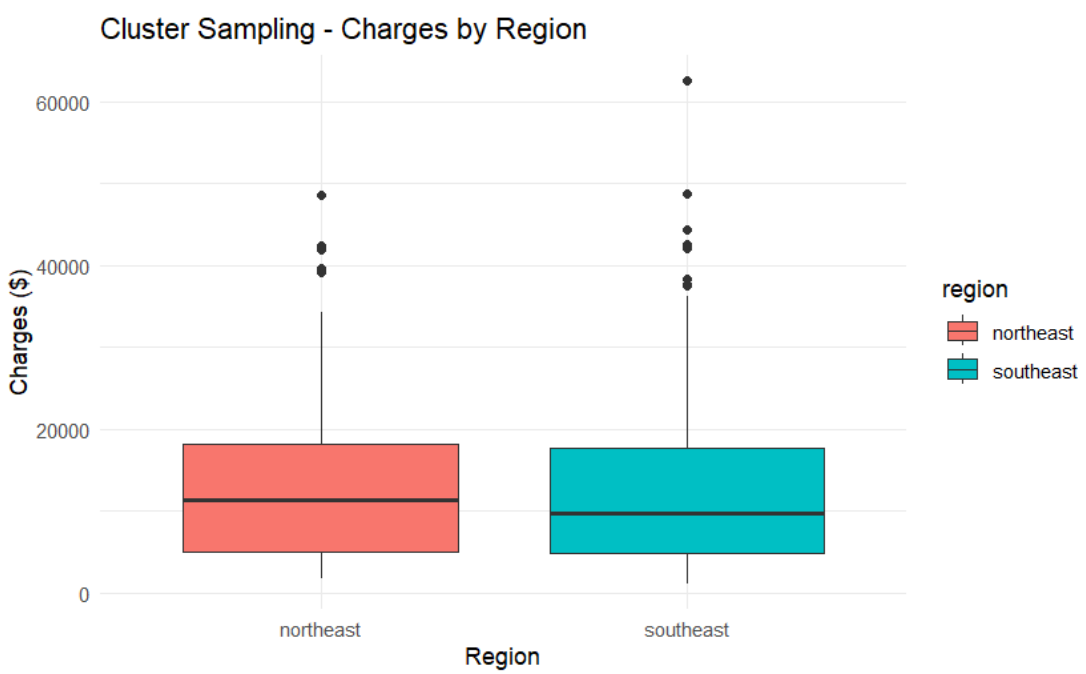
Graphical analysis is an important step in evaluating and comparing results obtained under different sampling designs. It helps to visually identify patterns, trends, and variations in the data. By using graphs such as histograms, boxplots, and bar charts, it becomes easier to observe the shape of distributions, differences between groups, and the presence of outliers. It also supports interpretation by providing a clear visual understanding of the relationships among variables and helps to determine which sampling design provides the most reliable and consistent estimates.



The histogram shows the distribution of health insurance charges obtained from the sample selected using Simple Random Sampling. From the plot, it is observed that the majority of individuals have lower insurance charges, mostly below $15,000, with the frequency gradually decreasing as the charges increase. The distribution is positively skewed, indicating that a smaller number of individuals have very high medical charges (above $30,000). These higher charges are likely due to individuals with higher health risks, such as smokers or those with higher BMI values.



From the plot, it is evident that smokers have significantly higher insurance charges compared to non-smokers. The median charge for smokers is much higher, and their interquartile range (IQR) is also wider, indicating greater variation in medical expenses among smokers. In contrast, non-smokers have lower and more consistent insurance charges, with a few outliers showing higher costs.



The plot can be seen that the distribution of charges is fairly similar between the two regions. Both clusters show a wide range of charges with several outliers representing individuals who incurred higher medical expenses. The median charges and the interquartile ranges (IQRs) are also quite close, indicating that regional differences have a relatively small effect on insurance charges compared to other factors such as smoking or BMI.

RESULTS OF THE STUDY

RESULTS OF SAMPLE ONE

- Charges Variable Under Each Design

Sampling design	Sample size	Mean	Standard error	Total	Standard error
Simple Random Sampling	594	13444.246	378.176	17988401.47	506000.06
Stratified Sampling	594	13610.681	207.033	18211090.88	277009.57
Cluster Sampling	600	13037.902	423.998	1269239757.83	39326556.05
Population		13270.42		17755825	

- Proportion For Smoker Variable Under Each Design

Sampling Design	Sample size	Proportions	Standard error
Simple Random Sampling	594	**smoker** Yes - 0.202 No - 0.798	0.012
Stratified Sampling	594	Yes - 0.205 No - 0.795	0
Cluster Sampling	600	Yes - 0.198 No - 0.802	0.012
Population		Yes - 0.205 No - 0.795	

RESULTS OF SAMPLE TWO

- Charges Variable Under Each Design

Sampling design	Sample size	Mean	Standard error	Total	Standard error
Simple Random Sampling	594	12610.376	343.91	16872683.03	460151.88
Stratified Sampling	594	13358.021	218.672	17873032.03	292583.23
Cluster Sampling	600	14170.913	533.99	1462438236.49	139870377.31
Population		13270.42		17755825	

- Proportion For Smoker Variable Under Each Design

Sampling Design	Sample size	Proportions	Standard error
Simple Random Sampling	594	**smoker** Yes - 0.178 No - 0.822	0.012
Stratified Sampling	594	Yes - 0.205 No - 0.795	0
Cluster Sampling	600	Yes - 0.233 No - 0.767	0.015
Population		Yes - 0.205 No - 0.795	

COMPARISON OF THE TWO SAMPLES

- Charges

Chargers		Mean	Total
SRS	Sample 1	13444.246	17988401.47
	Sample 2	12610.376	16872683.03
STRATIFIED	Sample 1	13610.681	18211090.88
	Sample 2	13358.021	17873032.03
CLUSTER	Sample 1	13037.902	1269239757.83
	Sample 2	14170.913	1462438236.49
POPULATION		13270.42	17755825

- Proportion for smoker

Smoker	SRS		Stratified		Cluster		Population
	Sample 1	Sample 2	Sample 1	Sample 2	Sample 1	Sample 2	
Yes	0.202	0.178	0.205	0.205	0.198	0.233	0.205
No	0.798	0.822	0.795	0.795	0.802	0.767	0.795

RATIO ESTIMATION

In this study, BMI (Body Mass Index) was used as the auxiliary variable, and charges were used as the study variable to perform ratio estimation. This selection is appropriate because there is a positive relationship between BMI and medical insurance charges individuals with higher BMI values often tend to have higher healthcare costs.

Sampling Design	Sample 1	Sample 2
Simple Random Sampling	449.7019	432.9663
Stratified Sampling	409.2237	436.3003
Cluster Sampling	469.7914	536.4435

CONCLUSION

This study analyzed the US Health Insurance Dataset containing 1,338 observations and several key variables such as age, BMI, charges, region, and smoker status. Three sampling designs Simple Random Sampling (SRS), Stratified Sampling, and Two-Stage Cluster Sampling were applied to estimate the population mean, total, standard error and proportion and to identify the most efficient sampling method.

From the analysis, it was observed that Stratified Sampling consistently produced estimates that were closest to the actual population values and had smaller standard errors compared to the other sampling methods. This indicates higher precision and reliability. The Simple Random Sampling method also provided reasonable estimates but showed slightly higher variability. In contrast, Cluster Sampling resulted in larger standard errors and greater deviations from the population values, mainly due to higher within-cluster homogeneity.

Therefore, based on the comparison of results from both samples, Stratified Sampling was identified as the most suitable sampling design for this dataset. It provided the most accurate and stable estimates by ensuring proportional representation of smokers and non-smokers, the key subgroups influencing medical insurance charges.

R CODES

```r
# PART 1: SETUP AND CENSUS CALCULATION

# 1. Install and Load Required Packages
install.packages("survey")
install.packages("readr")# Recommended package for reading CSV/Excel files
install.packages("dplyr")
library(survey) # Used for complex survey design analysis
setwd("C:/Users/HP/Documents")
library(readr)
library(dplyr)

# 2. Data Import (Assuming the CSV file is in the working directory)
# The file was named "insurance.csv"
file_path <- "insurance.csv"
census_data <- read_csv(file_path)


# Convert all character columns (categorical) to factors for proper analysis
census_data <- census_data %>%
  mutate(across(where(is.character), as.factor))
# structure check
str(census_data)

# If 'region' exists but is numeric, convert it to a labeled factor
if ("region" %in% names(census_data) && is.numeric(census_data$region)) {
  census_data$region <- factor(census_data$region,
                               levels = c(1, 2, 3, 4),
                               labels = c("northeast", "northwest", "southeast", "southwest"))
}
# 3. Population Size (N)
N <- nrow(census_data)
print(paste("Total Population Size (N):", N))

# 4. Define Quantitative and Categorical Variables
# These variables will be used by the analysis function for mean/proportion calculations.

#quantitative variables
quantitative_vars <- c("age" , "bmi" , "charges", "children")

```

```r

#Categorical variables
categorical_vars <- c("sex" ,"smoker" , "region")

# 5. Census (True Population) Values Calculation
pop <- list()

#6. Function to calculate true census mean and total for quantitative variables
for (var in quantitative_vars) {
  mean_val <- mean(census_data[[var]], na.rm = TRUE)
  total_val <- sum(census_data[[var]], na.rm = TRUE)
  pop[[var]] <- list(Mean = mean_val, Total = total_val)
}

#7.Categorical Variables: Proportion and Count of First Level
for (var in categorical_vars) {
  # Get the first level for the proportion calculation
  first_level <- levels(census_data[[var]])[1]

  # Calculate proportion of the first level
  proportion_val <- sum(census_data[[var]] == first_level, na.rm = TRUE)
  total_val <- sum(census_data[[var]] == first_level, na.rm = TRUE)

  pop[[var]] <- list(
    Level = first_level,
    Proportion = proportion_val,
    Total = total_val
  )
}

# 8. Display results
cat("\n Population (True Census) Estimates\n\n")

cat("$age\n")
cat("$age$Mean =", round(pop$age$Mean, 5), "\n")
cat("$age$Total =", round(pop$age$Total, 2), "\n\n")

```

```r
cat("$bmi\n")
cat("$bmi$Mean =", round(pop$bmi$Mean, 5), "\n")
cat("$bmi$Total =", round(pop$bmi$Total, 2), "\n\n")

cat("$children\n")
cat("$children$Mean =", round(pop$children$Mean, 5), "\n")
cat("$children$Total =", round(pop$children$Total, 2), "\n\n")

cat("$charges\n")
cat("$charges$Mean =", round(pop$charges$Mean, 5), "\n")
cat("$charges$Total =", round(pop$charges$Total, 2), "\n\n")

cat("$sex\n")
cat("$sex$Level =", pop$sex$Level, "\n")
cat("$sex$Proportion =", round(pop$sex$Proportion, 5), "\n")
cat("$sex$Total =", pop$sex$Total, "\n\n")

cat("$smoker\n")
cat("$smoker$Level =", pop$smoker$Level, "\n")
cat("$smoker$Proportion =", round(pop$smoker$Proportion, 5), "\n")
cat("$smoker$Total =", pop$smoker$Total, "\n\n")

cat("$region\n")
cat("$region$Level =", pop$region$Level, "\n")
cat("$region$Proportion =", round(pop$region$Proportion, 5), "\n")
cat("$region$Total =", pop$region$Total, "\n")

# PART 2: CORE ESTIMATION FUNCTION

# This function estimates mean, total, and proportion (and their SEs)
# for all quantitative and categorical variables under a given survey design.
# It works for your insurance dataset and matches your project's structure.
#Take a Simple Random Sample (SRS)

set.seed(123)
srs_sample <- census_data %>% sample_n(400)

```

```r

# Define a Survey Design Object
survey_design <- svydesign(ids = ~1, data = srs_sample, fpc = rep(N, nrow(srs_sample)))


#Define the Estimation Function
perform_survey_analysis <- function(survey_design, pop_total_N, quant_vars, cat_vars, design_name = "SRS") {

  results <- list()
  n_sample <- nrow(survey_design$variables)

  cat(paste0("\n--- ", design_name,
             " Estimates and SEs (Sample Size n = ", n_sample, ") ---\n"))
  cat(paste("Population Size (N):", pop_total_N, "\n\n"))

  #Quantitative Variables: Mean and Total
  for (var in quant_vars) {
    if (!var %in% names(survey_design$variables)) next

    formula_mean <- as.formula(paste("~", var))

    # Estimate Mean and SE
    mean_est <- svymean(formula_mean, survey_design, na.rm = TRUE)

    # Estimate Total and SE (scaled to population size)
    total_est <- svytotal(formula_mean, survey_design, na.rm = TRUE)

    # Store results in list
    results[[var]] <- list(
      Mean = as.numeric(coef(mean_est)),
      SE_Mean = as.numeric(SE(mean_est)),
      Total = as.numeric(coef(total_est)),
      SE_Total = as.numeric(SE(total_est))
    )
```

```r
    # Print the variable's results
    cat(paste0("Variable: ", var,
               " | Mean = ", round(results[[var]]$Mean, 3),
               " (SE = ", round(results[[var]]$SE_Mean, 3), ")\n"))
  }


  # Categorical Variables: Proportion (of first level) and Total
  for (var in cat_vars) {
    if (!var %in% names(survey_design$variables)) next

    first_level <- levels(survey_design$variables[[var]])[1]
    dummy_name <- paste0(var, "_dummy")

    # Create dummy variable (1 if first level, 0 otherwise)
    survey_design$variables[[dummy_name]] <- as.numeric(
      survey_design$variables[[var]] == first_level
    )

    formula_prop <- as.formula(paste("~", dummy_name))

    # Estimate Proportion (Mean of dummy) and SE
    prop_est <- svymean(formula_prop, survey_design, na.rm = TRUE)

    # Estimate Total and SE
    total_est <- svytotal(formula_prop, survey_design, na.rm = TRUE)

    # Store results
    results[[var]] <- list(
      Proportion_Level = first_level,
      Proportion = as.numeric(coef(prop_est)),
      SE_Proportion = as.numeric(SE(prop_est)),
      Total = as.numeric(coef(total_est)),
      SE_Total = as.numeric(SE(total_est))
    )

```

```r
    # Print results for categorical variable
    cat(paste0("Variable: ", var,
               " | Proportion of '", first_level, "' = ",
               round(results[[var]]$Proportion, 3),
               " (SE = ", round(results[[var]]$SE_Proportion, 3), ")\n"))
  }


  cat("\n End of ", design_name, " estimates.\n")
  return(results)
}


#Run the Estimation Function for Simple Random Sampling (SRS)
results_srs <- perform_survey_analysis(
  survey_design = survey_design,
  pop_total_N   = N,
  quant_vars    = quantitative_vars,
  cat_vars      = categorical_vars,
  design_name   = "Simple Random Sampling (SRS)"
)


# PART 3: SIMPLE RANDOM SAMPLING (SRS)
# 1. Decide Sample Size:
rsampcalc <- function(N, e = 3, ci = 95, p = 0.5) {
  z <- ifelse(ci == 95, 1.96, ifelse(ci == 99, 2.58, ifelse(ci == 90, 1.65, 1.96)))
  e <- e / 100
  n0 <- (z^2 * p * (1 - p)) / (e^2)
  n <- n0 / (1 + ((n0 - 1) / N))
  return(ceiling(n))
}


N <- nrow(census_data)
n_srs <- rsampcalc(N, e = 3, ci = 95)
cat("Calculated Sample Size (n_srs):", n_srs, "\n")
```

```r
# 2. Draw the SRS sample using base R sample_n
run_srs <- function(seed, label) {
  set.seed(seed) # Set seed for reproducibility
  srs_sample <- census_data %>%
    sample_n(size = min(n_srs, nrow(census_data)), replace = FALSE)

  # 3. Define the Survey Design (svydesign)
  # SRS does not strictly require the 'survey' package for estimates,
  # but defining a design allows for consistent use of svymean/svytotal.
  # We include the Finite Population Correction (fpc) which is necessary since N is known (census data).
  srs_design <- svydesign(
    ids = ~1,                # No clusters/PSUs
    fpc = rep(N, nrow(srs_sample)),     # Finite Population Correction (fpc = N, N, ...)
    data = srs_sample
  )
  # 4.Weight Calculation for BOTH SRS Samples
  srs_sample <- srs_sample %>%
    mutate(
      prob = n_srs / N,
      weight = 1 / prob
    )

  cat("\n", label, "\n")
  cat("SRS Sample Size =", nrow(srs_sample), "\n")
  cat("Weight Summary:\n")
  print(summary(srs_sample$weight))

  # 5. Perform Analysis using the custom function
  srs_result <- perform_survey_analysis(
    survey_design = srs_design,
    pop_total_N   = N,
    quant_vars    = quantitative_vars,
    cat_vars      = categorical_vars,
    design_name   = label
  )
```

```r
  return(srs_result)
}

#6. Run Two Independent SRS Samples
srs_result1 <- run_srs(101, "Simple Random Sample 1 (SRS)")
srs_result2 <- run_srs(202, "Simple Random Sample 2 (SRS)")

#7.  Compare Mean Estimates
print(" COMPARISON OF MEAN ESTIMATES ")
print(paste("Charges Mean - Sample 1:", round(srs_result1$charges$Mean, 2)))
print(paste("Charges Mean - Sample 2:", round(srs_result2$charges$Mean, 2)))
print(paste("BMI Mean - Sample 1:", round(srs_result1$bmi$Mean, 2)))
print(paste("BMI Mean - Sample 2:", round(srs_result2$bmi$Mean, 2)))

#8. Compare Proportion Estimates (e.g., Smokers)
print("=== COMPARISON OF PROPORTION ESTIMATES (Smoker) ===")
print(paste("Smoker Proportion - Sample 1:", round(srs_result1$smoker$Proportion, 3)))
print(paste("Smoker Proportion - Sample 2:", round(srs_result2$smoker$Proportion, 3)))
#9.Compare Total Estimates
print(" COMPARISON OF TOTAL ESTIMATES (Charges & BMI)")

print(paste("Charges Total - Sample 1:",
            round(srs_result1$charges$Total, 2)))
print(paste("Charges Total - Sample 2:",
            round(srs_result2$charges$Total, 2)))

print(paste("BMI Total - Sample 1:",
            round(srs_result1$bmi$Total, 2)))
print(paste("BMI Total - Sample 2:",
            round(srs_result2$bmi$Total, 2)))

# Compare Standard Errors of Total Estimates
print("STANDARD ERRORS OF TOTAL ESTIMATES (Charges & BMI)")

print(paste("Charges SE_Total - Sample 1:",
            round(srs_result1$charges$SE_Total, 2)))
print(paste("Charges SE_Total - Sample 2:",
            round(srs_result2$charges$SE_Total, 2)))
```

```r
print(paste("BMI SE_Total - Sample 1:",
            round(srs_result1$bmi$SE_Total, 2)))
print(paste("BMI SE_Total - Sample 2:",
            round(srs_result2$bmi$SE_Total, 2)))
# PART 4: STRATIFIED SAMPLING

# 1. Select Stratificatilon Variable: smoker
strat_var <- "smoker"

# Calculate Stratum Sizes
stratum_stats <- census_data %>%
  group_by(!!sym(strat_var)) %>%
  summarise(
    Nh = n(),
    Sh = sd(charges, na.rm = TRUE)
  )
# Total population and total sample size
N_total <- sum(stratum_stats$Nh)
n_strat <- rsampcalc(N_total, e = 3, ci = 95)
print(paste("Calculated Total Sample Size (n):", n_strat))

# Neyman Allocation Formula)
stratum_stats <- stratum_stats %>%
  mutate(
    weight = Nh * Sh,
    n_sample = round(n_strat * (weight / sum(weight)))
  )

# Adjust rounding (ensure total sample = n_strat)
diff <- n_strat - sum(stratum_stats$n_sample)
if (diff != 0) {
  max_row <- which.max(stratum_stats$n_sample)
  stratum_stats$n_sample[max_row] <- stratum_stats$n_sample[max_row] + diff
}

print(" Neyman Allocation Sample Sizes by Stratum")
print(stratum_stats)
```

```r
# 4. Draw the Stratified Sample
run_stratified_neyman <- function(seed, label) {
  set.seed(seed)
  strat_sample <- census_data %>%
    group_by(smoker) %>%
    group_modify(~sample_n(.x, size = stratum_stats$n_sample[stratum_stats$smoker == .y$smoker])) %>%
    ungroup()
  # Compute sampling weights
  strat_sample <- strat_sample %>%
    left_join(stratum_stats %>% select(smoker, Nh, n_sample), by = "smoker") %>%
    mutate(weight = Nh / n_sample)

  cat("Sample size (n):", nrow(strat_sample), "\n")
  cat("Weight summary:\n")

  print(paste("Sample size (n):", nrow(strat_sample)))
  print("Weight summary:")
  print(summary(strat_sample$weight))

  # 5. Define the Survey Design (svydesign)
  strat_design <- svydesign(
    ids = ~1,
    strata = ~smoker,                                    # Stratification variable
    fpc = ~Nh,                        # The total population size N in each stratum
    data = strat_sample
  )

  # 6. Perform Analysis
  result <- perform_survey_analysis(
    survey_design = strat_design,
    pop_total_N   = N,
    quant_vars    = quantitative_vars,
    cat_vars      = categorical_vars,
    design_name   = "Stratified Sampling"
  )
  return(result)
}
```

```r
#Run two independent stratified samples
strat_result1 <- run_stratified_neyman(101, "Stratified Sample 1 (Smoker)")
strat_result2 <- run_stratified_neyman(202, "Stratified Sample 2 (Smoker)")

#  Comparison of Mean Estimates
print("Comparison of Mean Estimates ")
print(paste("Sample 1 - Charges Mean:", round(strat_result1$charges$Mean, 2)))
print(paste("Sample 2 - Charges Mean:", round(strat_result2$charges$Mean, 2)))
print(paste("Sample 1 - BMI Mean:", round(strat_result1$bmi$Mean, 2)))
print(paste("Sample 2 - BMI Mean:", round(strat_result2$bmi$Mean, 2)))
#COMPARISON OF TOTAL ESTIMATES + STANDARD ERRORS
cat("\n COMPARISON OF TOTAL ESTIMATES (Charges & BMI) - Stratified Sampling \n")

results_strat_totals <- data.frame(
  Variable = c("charges", "charges", "bmi", "bmi"),
  Sample = c("Sample 1", "Sample 2", "Sample 1", "Sample 2"),
  Total_Estimate = c(
    round(strat_result1$charges$Total, 2),
    round(strat_result2$charges$Total, 2),
    round(strat_result1$bmi$Total, 2),
    round(strat_result2$bmi$Total, 2)
  ),
  SE_Total = c(
    round(strat_result1$charges$SE_Total, 2),
    round(strat_result2$charges$SE_Total, 2),
    round(strat_result1$bmi$SE_Total, 2),
    round(strat_result2$bmi$SE_Total, 2)
  )
)
print(results_strat_totals)

# PART 5: TWO-STAGE CLUSTER SAMPLING

# 1. Select Clustering Variable (PSU): region
cluster_var <- "region"
# 2. Decide Sample Size:
N <- nrow(census_data)
M <- length(unique(census_data[[cluster_var]]))
```

```r
M <- length(unique(census_data[[cluster_var]]))
n_total <- 600
avg_cluster_size <- N / M
n_clusters <- ceiling(M / 2)
n_sub_sample <- ceiling(n_total / n_clusters)
print(paste("Total clusters in population (M):", M))
print(paste("Selected clusters (m):", n_clusters))
print(paste("Expected individuals per cluster (k):", n_sub_sample))
print(paste("Expected total sample size (n):", n_clusters * n_sub_sample))

# 3. Define Cluster Frame (All unique region)
cluster_frame <- census_data %>% distinct(region) %>% pull(region)
M <- length(cluster_frame) # Total number of clusters in the population

# 4. Stage 1: Sample 'm' clusters (region)
run_cluster_sample <- function(seed, name) {
  set.seed(seed)
  sampled_clusters <- sample(cluster_frame, n_clusters)

  # 5. Stage 2: Sample individual within each selected cluster
  cluster_sample_data <- census_data %>%
    filter(region %in% sampled_clusters) %>%
    group_by(region) %>%
    sample_n(size = n_sub_sample, replace = FALSE) %>%
    ungroup()

  # Determine the total cluster size (Ni) for FPC in the second stage
  cluster_sizes_Ni <- census_data %>% count(region) %>% rename(Ni = n)
  cluster_sample_data <- left_join(cluster_sample_data, cluster_sizes_Ni, by = "region")

  # 6. Define the Survey Design (svydesign)
  ## Add FPC for each stage
  cluster_sample_data$FPC1 <- M
  cluster_sample_data$FPC2 <- cluster_sample_data$Ni
```

```r
  #  cluster sizes (Ni) for FPC in the second stage.
  cluster_design <- svydesign(
    ids = ~region,
    fpc = ~Ni,
    data = cluster_sample_data
  )


  # 7. Perform Analysis
  results <- perform_survey_analysis(
    survey_design = cluster_design,
    pop_total_N   = N,
    quant_vars    = quantitative_vars,
    cat_vars      = categorical_vars,
    design_name   = paste("Two-Stage Cluster Sampling -", name)
  )
  return(results)
}

# Run Two Independent Cluster Samples
cluster_result1 <- run_cluster_sample(101, "Cluster Sample 1")
cluster_result2 <- run_cluster_sample(202, "Cluster Sample 2")


print( "Comparison of Mean Estimates")
print(data.frame(
  Variable = c("charges", "bmi"),
  Sample1_Mean = c(round(cluster_result1$charges$Mean, 2),
                   round(cluster_result1$bmi$Mean, 2)),
  Sample2_Mean = c(round(cluster_result2$charges$Mean, 2),
                   round(cluster_result2$bmi$Mean, 2))
))

# COMPARISON OF TOTAL ESTIMATES + STANDARD ERRORS (Cluster Sampling)
cat("\n COMPARISON OF TOTAL ESTIMATES (Charges & BMI) - Cluster Sampling \n")

```

```r
results_cluster_totals <- data.frame(
  Variable = c("charges", "charges", "bmi", "bmi"),
  Sample = c("Sample 1", "Sample 2", "Sample 1", "Sample 2"),
  Total_Estimate = c(
    round(cluster_result1$charges$Total, 2),
    round(cluster_result2$charges$Total, 2),
    round(cluster_result1$bmi$Total, 2),
    round(cluster_result2$bmi$Total, 2)
  ),
  SE_Total = c(
    round(cluster_result1$charges$SE_Total, 2),
    round(cluster_result2$charges$SE_Total, 2),
    round(cluster_result1$bmi$SE_Total, 2),
    round(cluster_result2$bmi$SE_Total, 2)
  )
)

print(results_cluster_totals)


#  PART 6: Graphical analysis

install.packages("ggplot2")
library(ggplot2)

#Simple Random Sampling
set.seed(101)
srs_sample <- census_data %>%
  sample_n(300)

# Stratified Sample by Smoking Status
strat_sample <- census_data %>%
  group_by(smoker) %>%
  sample_n(100) %>%
  ungroup()
```

```r
# Cluster Sample by Region
selected_regions <- sample(unique(census_data$region), 2)
cluster_sample <- census_data %>%
  filter(region %in% selected_regions) %>%
  group_by(region) %>%
  sample_n(100) %>%
  ungroup()


#Plot 1: SRS - Histogram of Charges
ggplot(srs_sample, aes(x = charges)) +
  geom_histogram(binwidth = 2000, fill = "#1f77b4", color = "black") +
  labs(title = "SRS Sampling - Charges Distribution",
       x = "Charges ($)", y = "Frequency") +
  theme_minimal()


#Plot 2: Stratified - Boxplot of Charges by Smoking Status
ggplot(strat_sample, aes(x = smoker, y = charges, fill = smoker)) +
  geom_boxplot() +
  labs(title = "Stratified Sampling - Charges by Smoking Status",
       x = "Smoker", y = "Charges ($)") +
  theme_minimal()


# Plot 3: Cluster Sampling - Charges by Region
ggplot(cluster_sample, aes(x = region, y = charges, fill = region)) +
  geom_boxplot() +
  labs(title = "Cluster Sampling - Charges by Region",
       x = "Region", y = "Charges ($)") +
  theme_minimal()


# Population distribution by region
region_dist <- census_data %>%
  count(region) %>%
  mutate(Proportion = n / sum(n))
```

```r
print(region_dist)


#PART 7: RATIO ESTIMATORS

#Keep only numeric variables needed for ratio estimation
census_data$bmi <- as.numeric(census_data$bmi)
census_data$charges <- as.numeric(census_data$charges)

#Population values
N <- nrow(census_data)
pop_mean_bmi <- mean(census_data$bmi)
pop_total_bmi <- sum(census_data$bmi)
# Ratio Estimator Function
ratio_estimator <- function(sample_data, y_var = "charges", x_var = "bmi", pop_mean_x) {
  sample_data[[x_var]] <- as.numeric(sample_data[[x_var]])
  sample_data[[y_var]] <- as.numeric(sample_data[[y_var]])

  xbar <- mean(sample_data[[x_var]])
  ybar <- mean(sample_data[[y_var]])

  R_hat <- ybar / xbar
  Mean_Y_hat <- R_hat * pop_mean_x

  return(list(Ratio = R_hat, Est_Mean_Charges = Mean_Y_hat))
}

#PART 1 – SIMPLE RANDOM SAMPLING (SRS)

run_srs_ratio <- function(seed, label) {
  set.seed(seed)
  sample_srs <- census_data %>% sample_n(200)
```

```r

  res <- ratio_estimator(sample_srs, "charges", "bmi", pop_mean_bmi)

  return(data.frame(
    Sample = label,
    Ratio_Estimate = round(res$Ratio, 4),
    Estimated_Mean_Charges = round(res$Est_Mean_Charges, 2)
  ))
}

srs1 <- run_srs_ratio(101, "SRS Sample 1")
srs2 <- run_srs_ratio(202, "SRS Sample 2")

# PART 2 - STRATIFIED SAMPLING (by smoker-10% from each stratum)

run_strat_ratio <- function(seed, label) {
  set.seed(seed)
  sample_strat <- census_data %>%
    group_by(smoker) %>%
    sample_frac(0.10) %>%
    ungroup()

  res <- ratio_estimator(sample_strat, "charges", "bmi", pop_mean_bmi)

  return(data.frame(
    Sample = label,
    Ratio_Estimate = round(res$Ratio, 4),
    Estimated_Mean_Charges = round(res$Est_Mean_Charges, 2)
  ))
}

strat1 <- run_strat_ratio(303, "Stratified Sample 1")
strat2 <- run_strat_ratio(404, "Stratified Sample 2")

```

```r
#PART 3 - TWO-STAGE CLUSTER SAMPLING (by region)

run_cluster_ratio <- function(seed, label) {
  set.seed(seed)

  # Stage 1: select 2 clusters
  selected_clusters <- sample(unique(census_data$region), 2)

  # Stage 2: sample equal number per cluster
  sample_cluster <- census_data %>%
    filter(region %in% selected_clusters) %>%
    group_by(region) %>%
    sample_n(50) %>%  # 50 from each selected region
    ungroup()

  res <- ratio_estimator(sample_cluster, "charges", "bmi", pop_mean_bmi)

  return(data.frame(
    Sample = label,
    Ratio_Estimate = round(res$Ratio, 4),
    Estimated_Mean_Charges = round(res$Est_Mean_Charges, 2)
  ))
}

cluster1 <- run_cluster_ratio(505, "Cluster Sample 1")
cluster2 <- run_cluster_ratio(606, "Cluster Sample 2")

#Combine & Display Final Ratio Estimation Results

ratio_results <- rbind(srs1, srs2, strat1, strat2, cluster1, cluster2)
print(ratio_results)
#TRUE POPULATION MEAN OF CHARGES
pop_mean_charges <- mean(census_data$charges, na.rm = TRUE)

#Add Population Mean Column to Ratio Table
ratio_results$Population_Mean <- round(pop_mean_charges, 4)
```

```
colnames(ratio_results) <- c("Sampling_Design", "Ratio_Estimate",
                             "Estimated_Mean_Charges", "Population_Mean")

#Display result
print(ratio_results)
```

Data set:

https://www.kaggle.com/datasets/teertha/ushealthinsurancedataset

Thank You.